

Build a simple low power DC-DC Buck Converter

Introduction to the course:	In applications such as power supplies, battery-powered systems, LED drivers, and motor controllers, it is often necessary to step down a higher DC voltage to a lower DC voltage. A buck converter provides this functionality with high efficiency compared to linear regulators. This project intends to provide a hands-on approach to build a low-power DC-DC buck converter.
Contents	• To introduce circuit simulation using LT-spice software. • To create a dc-dc buck converter circuit using breadboard and test with a load(fan).
Methodology	• Step 1 - Circuit Simulation of the dc-dc buck converter using LT-spice. • Step 2 - Actual component testing on breadboard. With the required input supply, the output results are monitored using multimeter.
Course Prerequisites	• Understanding of buck converter operations • Basics of RLC circuits

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Part 1 – Circuit Simulation using LT-Spice

Part 2 – Physical design of the circuit

Build a simple low power DC-DC Buck Converter using LT-Spice

Part – 1 Circuit Simulation using LT-Spice

Aim

Basic understanding about LT-Spice

LT-Spice simulation installation instructions

Connections - Circuit Diagram - Detailed Steps

Task - Exercise

Fundamental learning

Topic	Resource
Concept of Buck-Converter	YouTube link
Components & Working of Dc-Dc Buck converter	YouTube link

Aim

Very often a low power buck converter used to convert some high voltage to a lower voltage is required in a system. This build exercise is to learn to build one such simple converter, using LT-Spice simulation in an open loop.

LT-Spice - Introduction

LT-Spice is an open source, high-performance simulation tool developed by Analog Devices. It is used to simulate, design, and test electronic circuits before physically building them.

- **Why LT-Spice is important?**

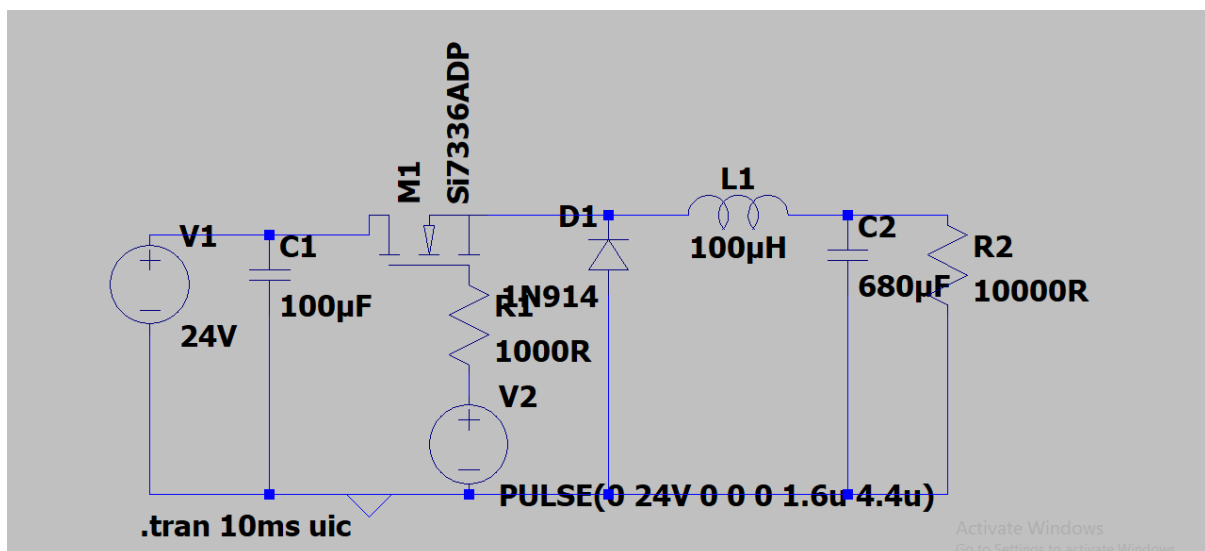
LT-Spice provides a safe learning environment where peoples can freely experiment with electronic circuits without risk of damaging physical components. It helps them to understand circuit behaviour by allowing real time visualization of voltage, current, and other parameters. By simulating before building, Peoples save time and cost as they as they do not need to buy multiple components or PCBs initially. LT-Spice also encourages peoples to develop strong problem -solving skills, teaching them how to identify and fix circuit issues logically. It offers valuable hands-on simulation experience that connects theoretical concepts learned in classrooms with practical real-world electronics. Most importantly, learning LT-Spice gives peoples and industry-relevant skill, preparing them for careers in research, development, and engineering industries.

LT-Spice Simulation Installation Instructions

- 1) **Open your web browser** (Such as Chrome, Firefox, or Edge).
- 2) In the search bar, type **LT-Spice** and press Enter.
- 3) Click on the official link from **Analog devices**, usually titled something like “LTspice” | **Analog devices**”. Or use this hyperlink to download [LT-Spice](#).
- 4) **On the LTspice page**, look for a **Download** button or section.
- 5) **Choose the correct version** for your operating system (Windows, macOS, etc.).
- 6) **Download the installer file** to your computer.
- 7) **Once the download is complete**, open the installer file.
- 8) **Follow the on-screen instructions** to install LT-Spice:
 - Accept the License Agreement
 - Choose the installation folder (or use the default)
 - Complete the installation process.
- 9) **After installation, open LT-Spice** from your desktop or Start Menu.
- 10) **Now are you ready to start using LT-Spice for circuit simulation.**

Connections

Circuit Diagram:



Detailed Connection Steps

Step 1: Open the **LT-Spice** simulator in your laptop.

Step 2: Create a new schematic

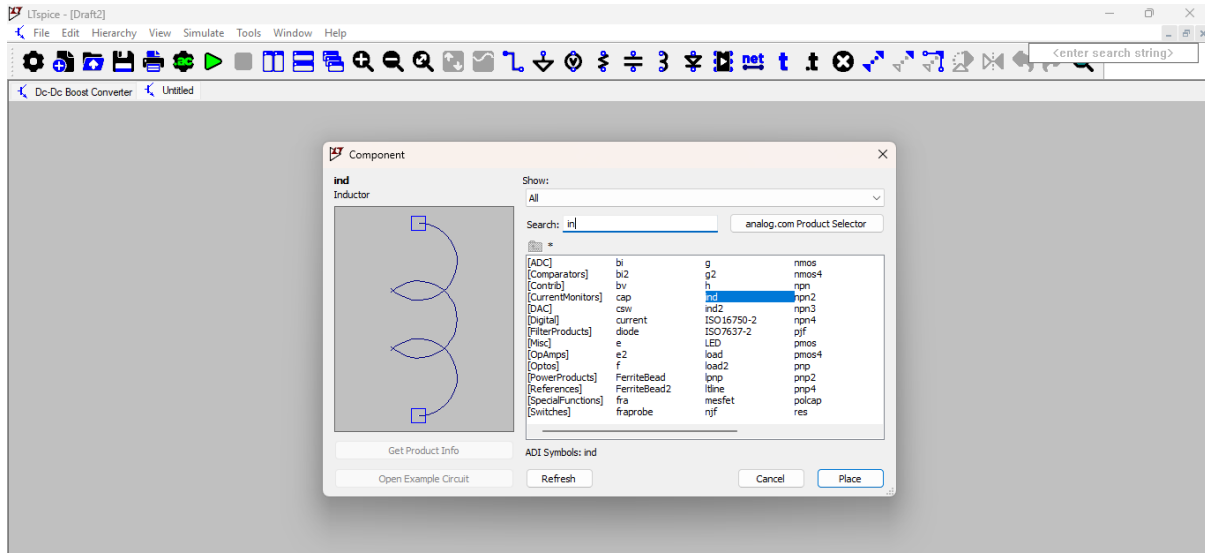
- Go to the **File** menu.
- Click on **New Schematic**.

Step 3: Place components

We need to place these components one by one:

Component	Symbol in LT-Spice	How to place
Voltage Source	Voltage (V)	Press F2, or Click  this icon then search for voltage
Inductor	Inductor (L)	Press F2, or click that icon and search ind
Diode	Diode(D)	Press F2, or click that icon and search diode
MOSFET	MOSFET (M)	Press F2, or click that icon and search nmos
Capacitor	Capacitor (C)	Press F2, or click that icon and search cap
Resistor	Resistor (R)	Press F2, or click that icon and search res
Ground	GND	Press F2, or click that icon and search gnd

Note:  Click this Icon in your New schematic and search all the respective components as mentioned in the above table.



Note: After selecting a component, **click** on the schematic area to **place it**.

Step 4: Place Two Voltage Sources

- **V1: (Input 5V DC Supply)**
- **V3: (PWM gate signal for MOSFET)**

Note: In the physical hardware setup, we are using a switching regulator IC – LM2575-ADJ. This IC contains an internal MOSFET, oscillator, and control circuitry required for buck conversion. By adding only a few external components such as an inductor, diode, and capacitors, it can efficiently step down a higher DC voltage to a desired lower DC voltage.

- **After placing the voltage sources, you can right-click on them to rename:**
- **Name the first voltage source as V1** this will be the **input DC supply** (e.g., 24V).
- **Name the second voltage source as V2** this will be the **PWM control source** for the MOSFET gate.

Step 5: Arrange Components Properly

Arrange the components based on the buck converter circuit:

1. V1 (input supply) connect to capacitor C1 and then to Drain (D) of MOSFET (M1).
2. Source (S) of MOSFET (M1) connects to ground (GND).
3. Drain (D) of MOSFET (M1) also connects to one side of inductor (L1).
4. Other side of L1 connects to:
 - Positive terminal of C2
 - One side of R2 (load)
 - Cathode (K) of diode D1

5. Anode (A) of D1 connects back to ground.
6. Negative side of capacitor C2 and other side of load resistor R2 connect to ground.
7. Gate (G) of MOSFET (M1) connects to V2 (PWM voltage source).

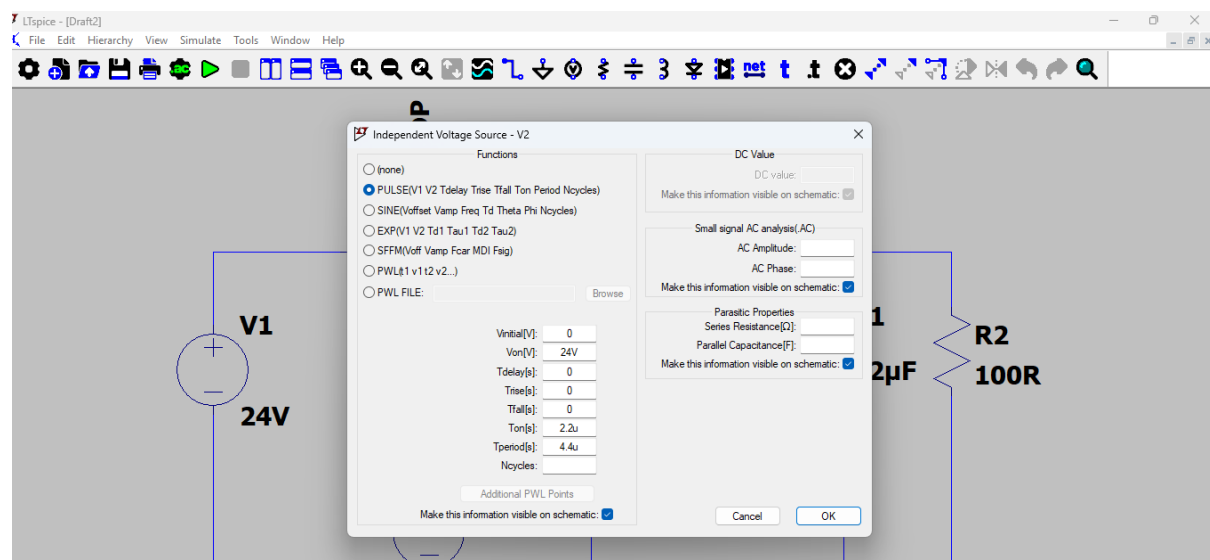
Step 6: Connect Wires

- Press F3 to start wiring.
- Connect each pin exactly as per the circuit diagram.
- Double check all grounds are properly connected.

Step 7: Component Values

Component	Value to set
V1 (voltage source)	DC = 24V
V3 (PWM source)	Will set as pulse source (explained below)
L1 (Inductor)	100 μH
D1(Diode)	Schottky diode (e.g., “ 1N5822 ”)
M1 (MOSFET)	Choose “ Si7336ADP ” or any standard NMOS
C1 (Input Capacitor)	100μF
C2 (Output Capacitor)	680μF
R1 (Resistor)	1000Ω(1K)
R2 (Resistor)	10000Ω(10K)

Note: This will generate a PWM square wave of about 50KHz frequency



Note: Use the above image for your references.

Understanding about Duty Cycle:

The Duty Cycle (%) tells us how long the MOSFET stays ON during one complete switching cycle.

Duty Cycle Formula:

$$\text{Duty Cycle(\%)} = \left(\frac{T_{on}}{T_{period}} \right) \times 100$$

$$\text{Duty Cycle} = (22/44) \times 100 = 50\%$$

Note: If you want to increase the Duty Cycle then increase the Ton period and observe the output.

Step 8: Add Ground Connections

Every voltage source and component must have a ground reference

- Press F2, type GND, place the ground symbol.
- Then wire the grounds as follows:
- Connections GND to:
 1. Connect GND to the negative terminal of V1 (input supply, 24V).
 2. Connect GND to the negative terminal of V2 (PWM source).
 3. Connect GND to the negative side of C1 (input capacitor).
 4. Connect GND to the source pin of MOSFET (M1).
 5. Connect GND to the negative terminal of C2 (output capacitor) and also the other side of R2 (load resistor).
 6. Connect GND to the anode of diode (D1).

Step 10: Setup Simulation Command

- Press S or go to simulate > Edit simulation Cmd.
- Choose Transient tab.
- Set stop time:
 - 1) Stop time: 10ms

- Click Ok.
- Place the. **tran 10ms uic** command on the schematic.

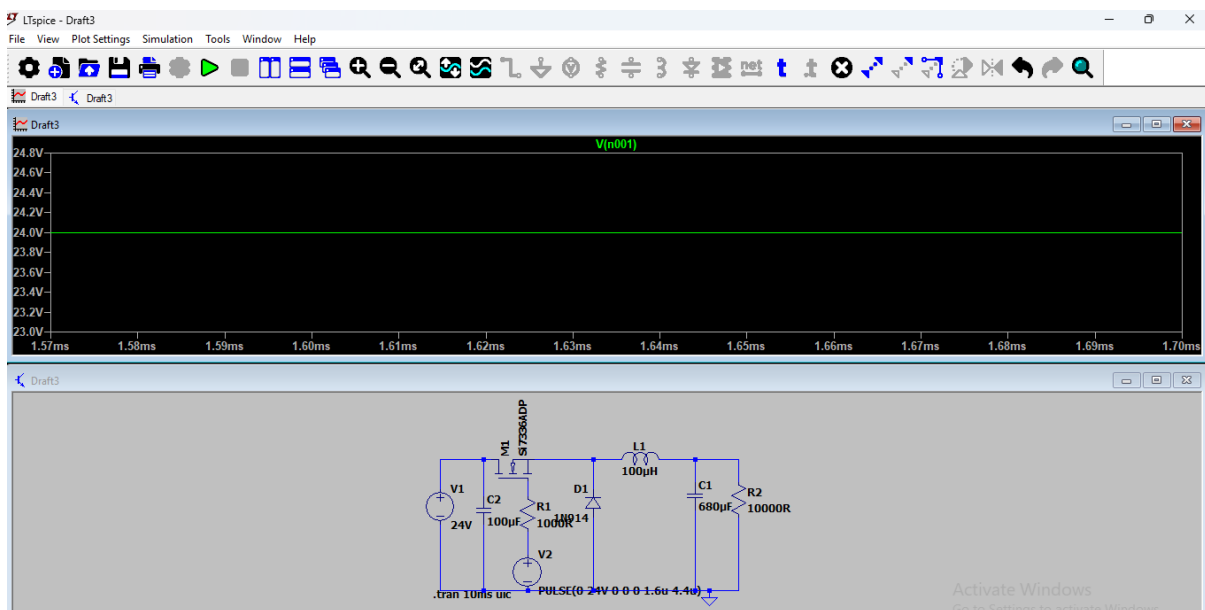
Step 11: Save and Run Simulation

- Click save.
- Then click the Run button.
- The simulation will start.
- You can probe:
 - 1) Voltage across the capacitor (C1).
 - 2) Current through Inductor (L1).
 - 3) Gate signal applied to MOSFET.

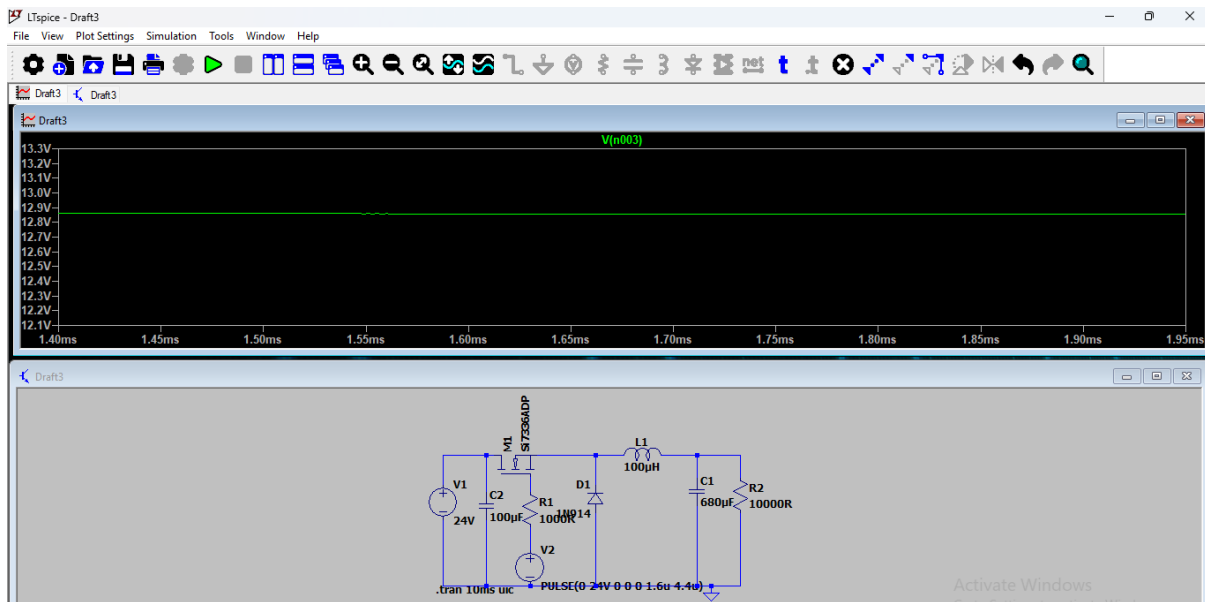
Note: Simulation video for your reference [\(link\)](#)

Expected Output:

Input Voltage: 24V



Output Voltage: 12V



Task:

- 1) Set the input voltage as 48V and design a Buck Converter to get an output of around 24V.
- 2) Simulate the same Buck Converter circuit with different load resistances (e.g., 60Ω, 120Ω, 180Ω) and observe how the output voltage changes.
- 3) Change the inductor value (e.g., 608µH to 300µH or 900µH) and observe how it affects the ripple in output voltage and current.
- 4) Increase or decrease the switching frequency and note how the output voltage ripple and efficiency are affected.
- 5) Plot and observe the Inductor Current, Capacitor Voltage, and MOSFET Drain-Source Voltage during simulation.

Build a simple low power DC-DC Buck Converter using physical circuit

Part 2 – Physical design of the circuit

Aim

Components

Connections - Circuit Diagram

Detailed Steps

Task - Exercise

Aim

Very often a low power buck converter used to convert some high voltage to a low voltage is required in a system. This build exercise is to learn to build one such simple converter, using an open loop.

Components

1. Buck IC (LM2575T ADJ) – 1
2. Electrolytic Capacitor (100uF 63V) - 1
3. Electrolytic Capacitor (470uF 63V) - 1
4. Schottky diode (1N4007) – 1
5. Inductor (330 uH 1A) – 1
6. Resistor (10K) – 1
7. Resistor (1K) – 1
8. Bread board – 1
9. Digital Multi meter - 1
- 10 Male to Male Jumper wire – 7
- 11 24V 2A Power adaptor – 1
- 12 Power Socket – 1
- 13 DC load (12V Fan) - 1
- 14 Oscilloscope – 1 (Optional)

Note: All the components are reusable



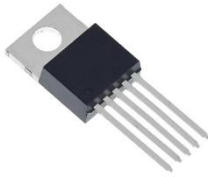
Capacitor



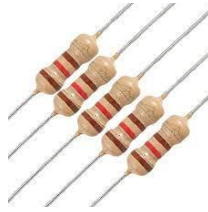
Inductor



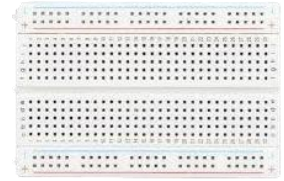
Diode



Buck IC



Resistors



Breadboard



Jumper Wires



DC load (Fan)



24V Power Adaptor

Introduction and Principle of Operation (Theory)

The buck converter, also referred to as a step-down converter, is a popular topology in power electronics that converts a higher input voltage to a lower output voltage. It is crucial in various applications, from portable devices to automotive systems, where specific components or subsystems require a lower voltage level to operate. The primary advantage of the buck converter is its simplicity, which enables efficient voltage conversion using a relatively small number of components.

The operating principle of the buck converter involves controlled energy transfer from the input to the output through switches, an inductor, and a capacitor. A high-side switch (usually a MOSFET) and a low-side switch (typically a diode) are employed in the buck converter to control the current flow through the inductor. By adjusting the duty cycle of the high-side switch, the average output voltage can be regulated proportionally to the input voltage.

When the high-side switch of a buck converter is switched on, it allows current to flow through the inductor, which stores energy in its magnetic field. This stored energy is then transferred to the output, charging the output capacitor and powering the load. When the high-side switch is turned off and the low-side switch is turned on, the inductor's magnetic field collapses, releasing the stored energy and maintaining the current flow to the load. The buck converter is designed to operate within a closed-loop control system, where a feedback mechanism continuously compares the output

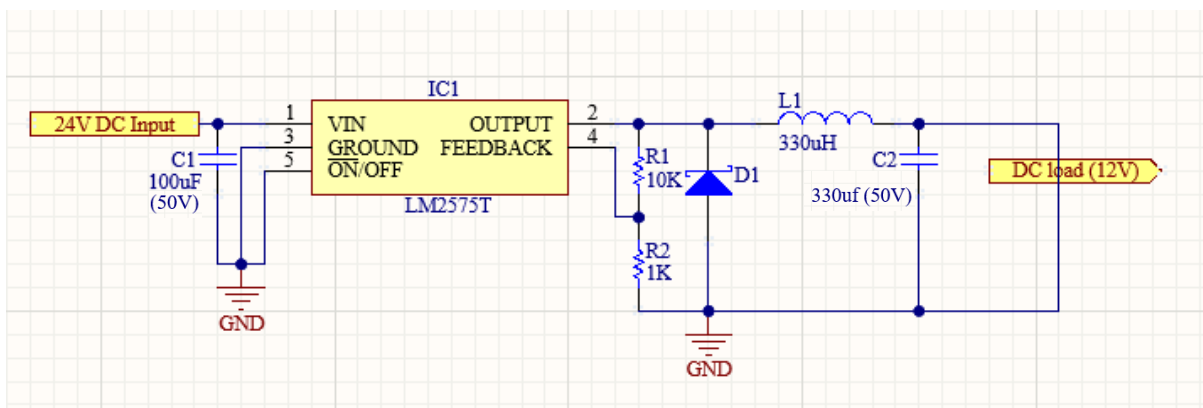
voltage to a reference voltage to ensure that the output voltage remains stable and regulated, regardless of changes in input voltage or load conditions.

Topic	Resource
Detailed principal of operation	link
Buck topology	link

Note: This DC-DC Buck converter project was implementing using LM2575-ADJ IC. For detailed information about the IC use this [reference link](#)

Connections

Circuit Diagram:



NOTE: First make the circuit diagram in the Bread board and verify the connection properly once you got the output then only go for soldering. Do not touch the tip of the soldering iron, its temperature can be as high as 380°C and can cause severe burns. Keep the cleaning sponge wet when soldering.

Detailed Connection Steps

Step-by-Step Procedure for LM2575 Buck Converter Circuit (24V to 12V)

Step 1: Place Components on Bread Board

- Place the **LM2575T IC** at the center of the board.
- Keep input capacitor (C1 = 100 μ F) near the VIN (pin 1) of LM2575.
- Keep output capacitor (C2 = 680 μ F) close to the output (pin 2) of LM2575.
- Place inductor (L1 = 330 μ H) next to the IC output (pin 2).
- Place diode (D1 – Schottky diode) near the IC output (pin 2).
- Place feedback resistors (R1 = 10k Ω , R2 = 1k Ω) near the Feedback (pin 4).

Step 2: Connect the Input Power Supply

- Connect **24V DC input supply** positive terminal to **Pin 1 (VIN)** of LM2575 through C1 (100 μ F).
- Connect the negative terminal of the power supply to **GND (Pin 3)**.

Step 3: Connect the Inductor & Diode

- Connect **Pin 2 (OUTPUT)** of LM2575 to one end of **L1 (330 μ H inductor)**.
- Connect the other end of **L1** to the positive side of **C2 (680 μ F)** and to the load.
- Connect a **Schottky diode (D1)**:
 - **Cathode (K)** to LM2575 Pin 2 (OUTPUT).
 - **Anode (A)** to GND.

Step 4: Connect the Output Capacitor & Load

- Connect **C2 (680 μ F capacitor)** across the load:
 - Positive terminal to Inductor output.
 - Negative terminal to GND.
- Connect your **12V DC load** (fan, etc.) across capacitor C2.

Step 5: Connect the Feedback Resistors

- Connect **R1 (10k Ω)** between Output (Pin 4 FEEDBACK) and Output line (after L1).
- Connect **R2 (1k Ω)** between Pin 4 (FEEDBACK) and GND.
- This resistor divider sets the output voltage (~12V).

Step 6: ON/OFF Pin Connection

- Connect **Pin 5 (ON/OFF)** to GND for normal operation.

Step 7: Double Check the Circuit

- Verify all **polarities** of capacitors and diode.
- Ensure correct resistor values for voltage setting.
- Make sure inductor is rated for required current ($\geq$ load current).

Step 8: Power Up the Circuit

- Apply 24V DC input at the input terminals.
- Use a digital multimeter to measure the output voltage across capacitor C2.
- You should read close to 12V DC (depending on resistor ratio).

Step 9: Test with Load

- Connect the DC load across the output capacitor (C2).
- Observe if the load is working properly at 12V.

Tasks:

1. Set the input voltage to 18V, 24V, and 28V using a DC power supply and measure the output voltage in each case to check whether the converter maintains $\sim 12V$ regulation.
2. Change the feedback resistor ratio (R1 and R2) to set different output voltages such as 5V, 9V, and 15V, then measure the actual outputs and compare with the theoretical calculation using $V_{out} = 1.23 \times (1 + R1/R2)$.
3. Replace the output capacitor with values of 220 μF , 470 μF , and 680 μF one by one and measure the output ripple voltage using an oscilloscope to see how capacitor size affects stability and filtering.
4. Use an oscilloscope to observe the inductor current waveform, the capacitor voltage, and the diode conduction behaviour, and record the measured ripple voltage and current.